

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

Random Variables and Expectations

Chapter 4 — Distributions, Expectation, and the Weak Law of Large Numbers

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Chapter Outline | What we will cover

Random Variables & Distributions
Joint, Marginal & Conditional Distributions
Continuous Random Variables
Expectation

Variance & Covariance
The Weak Law of Large Numbers
Applications
Summary

Learning Objectives

- ▶ Define **random variables** and their **PMF/CDF**
- ▶ Work with **joint**, **marginal**, and **conditional** distributions
- ▶ Compute **expectation**, **variance**, and **covariance**
- ▶ Use the **Weak Law of Large Numbers** to justify averaging

Section 1

Random Variables & Distributions

Definition 4.1

A **discrete random variable** X is a function mapping each outcome $\omega \in \Omega$ to a number $X(\omega)$.

Numbers from coins

Flip 3 fair coins; let X = number of heads.

$$X \in \{0, 1, 2, 3\}$$

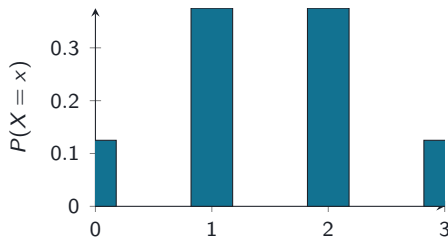
PMF and CDF | Two equivalent descriptions of a distribution

Definition 4.2: PMF

$p(x) = P(X = x)$, with $\sum_x p(x) = 1$.

Definition 4.3: CDF

$F(x) = P(X \leq x) = \sum_{u \leq x} p(u)$



PMF of $X = \text{\#heads in 3 flips}$

Section 2

Joint, Marginal & Conditional Distributions

Definition 4.4

$p(x, y) = P(X = x, Y = y)$, jointly describing X and Y .

Definition 4.6: Marginal

$$p(x) = \sum_y p(x, y)$$

Sums and differences of dice

Roll two dice; S = sum, D = difference. Their joint distribution $p(s, d)$ captures every combination.

Independence of Random Variables | S and D : independent or not?

Definition 4.7

X, Y are independent iff

$$p(x, y) = p(x) p(y) \quad \forall x, y$$

Checking S and D

$P(S = 2, D = 5) = 0$ but $P(S = 2) > 0$ and $P(D = 5) > 0$ — so $p(s, d) \neq p(s)p(d)$: **not independent**.

Bayes' rule for random variables (Def. 4.5)

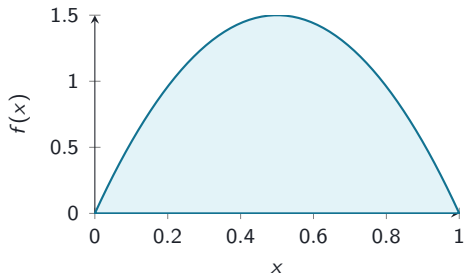
$$p(x | y) = \frac{p(y | x) p(x)}{p(y)}$$

Section 3

Continuous Random Variables

Properties of a PDF

- ▶ $f(x) \geq 0$ for all x
- ▶ $\int_{-\infty}^{\infty} f(x) dx = 1$
- ▶ $P(a \leq X \leq b) = \int_a^b f(x) dx$



A PDF can exceed 1

Unlike a PMF, $f(x)$ is a *density*, not a probability — it can be larger than 1 as long as the total area is 1.

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Section 4

Expectation

Expected Value | The probability-weighted average

Discrete

$$E[X] = \sum_x x p(x)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

A fair die roll

$$E[X] = \sum_{i=1}^6 i \cdot \frac{1}{6} = 3.5$$

Note $E[X]$ need not be a possible outcome.

Properties of Expectation | Linearity — the single most useful fact in probability

Linearity

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

Holds *whether or not* X, Y are independent.

Products need independence

If X, Y independent:

$$E[XY] = E[X] E[Y]$$

Section 5

Variance & Covariance

Mean, Variance, and Covariance | Location, spread, and joint spread

Quantity	Formula
Mean	$\mu = E[X]$
Variance	$\text{var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$
Covariance	$\text{cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

Key properties

$\text{var}(aX + b) = a^2 \text{var}(X)$. If X, Y independent, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$.

Section 6

The Weak Law of Large Numbers

IID Samples & the Sample Mean | Averaging many independent copies

IID

$X_1, \dots, X_n$ are **independent and identically distributed** — same distribution, mutually independent.

Sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad E[\bar{X}_n] = \mu$$

and $\text{var}(\bar{X}_n) = \sigma^2/n$ — shrinks with n .

Markov's and Chebyshev's Inequalities

| Bounding tail probabilities without knowing the shape

Markov's inequality

For $X \geq 0$ and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev's inequality

Apply Markov to $(X - \mu)^2$:

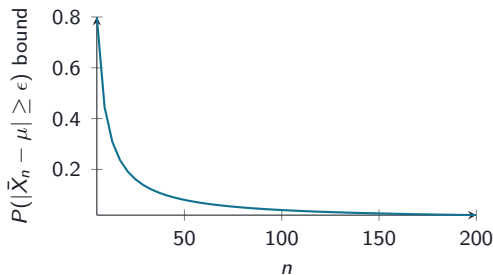
$$P(|X - \mu| \geq a) \leq \frac{\text{var}(X)}{a^2}$$

The Weak Law of Large Numbers | The sample mean concentrates around μ

Statement

For any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \xrightarrow{n \rightarrow \infty} 0$$



Bound shrinks as n grows



Section 7

Applications

Should You Accept a Bet? | Expected value as a decision rule

Simple bet

Pay \$1; win \$3 with probability $1/4$, else lose the dollar.

$$E[\text{payoff}] = \frac{1}{4}(3) + \frac{3}{4}(-1) = 0$$

Interpretation

A fair bet ($E = 0$) is break-even *on average* — but a single play is still risky. Casinos set $E < 0$ for the house edge.

Odds vs. probability

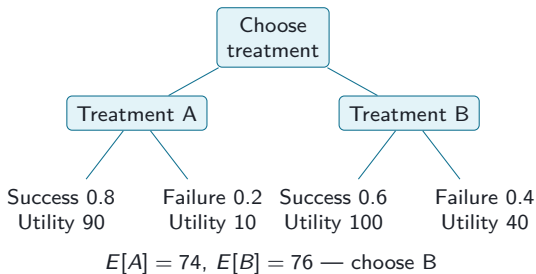
"3-to-1 odds" against an event means implied $P(\text{event}) = 1/(3 + 1) = 0.25$.

The bookmaker's edge

Real odds are set so the implied probabilities of all outcomes sum to *more than 1* — guaranteeing negative expected value for the bettor.

Decision Trees | Structuring a decision under uncertainty

Example: choosing a medical treatment. Compute the **expected utility** of each branch and pick the best.



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Section 8 Summary

Chapter Summary | Key takeaways

Describe

Random variables turn outcomes into numbers; PMFs/PDFs and CDFs describe their distribution.

Summarize

$E[X]$ (location), $\text{var}(X)$ (spread), $\text{cov}(X, Y)$ (joint spread) — linearity of $E[\cdot]$ is your best friend.

Why it matters

The Weak Law justifies using sample averages to estimate unknown means — the foundation of statistical inference.

Looking ahead

Ch. 5: the most useful named **probability distributions**.

Thank you!

Questions & Discussion

Next: Chapter 5 — Useful Probability Distributions